

5th International Conference on Industry 4.0 and Smart Manufacturing

Telematics and machine learning system for estimating the load condition of a heavy-duty vehicle

Fabio Ribeiro von Glehn^{a,*}, Bruno Henrique Pereira Gonçalves^b,
Marlize Garcia Fagundes Neto^a, João Paulo da Silva Fonseca^a

^aEscola de Engenharia Elétrica, Mecânica e de Computação - Universidade Federal de Goiás, Ingá Ave., Goiânia 74690-900, Brazil

^bGrupo Masut, Gerência de Pesquisa e Desenvolvimento – Anápolis – Goiás - Brazil

Abstract

This study employs machine learning techniques to determine whether trucks are loaded or empty based solely on data from the vehicle's communication network, especially from the engine module. We achieved an accuracy of over 85% for short hauls (0.5 to 5 km) and nearly 95% for long hauls (5 to 500 km). This approach not only streamlines fleet management, reducing the need for communication between managers and drivers but also plays a crucial role in research related to fuel consumption reduction and intelligent fault diagnostics. Minimizing dependence on imprecise and time-delayed human responses opens new avenues for effective studies and innovative solutions in the transportation industry.

© 2024 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 5th International Conference on Industry 4.0 and Smart Manufacturing

Keywords: Telematics; Heavy-duty vehicles; Fuel saving; Big data analytics; Machine learning

1. Introduction

In 2021, the International Energy Agency (IEA) published an overview of energy consumption in freight transport in Brazil. The IEA study shows that heavy trucks consume about 60% of the energy in freight transport in road, rail, and water modes in Brazil and, reports highlight that energy efficiency is a critical tool to reduce carbon emissions and air pollution from the transport sector and is part of a package of sustainability [1].

Reducing energy consumption in freight transport modes can have positive impacts on both the economic and environmental dimensions. However, studies aiming at energy reduction in road freight transport rely on the global

* Corresponding author. E-mail address: fabiooglehn@discente.ufg.br

understanding of traffic and driving characteristics. In this regard, vehicle telematics becomes relevant as part of the solution to this demand. Vehicle telematics is the technology that collects and transmits data from vehicles, such as speed, location, acceleration, braking, and idling [2]. Vehicle telematics can provide valuable insights into drivers' driving behavior and performance, as well as the condition and maintenance of vehicles [3]. Vehicle telematics can also help to improve the energy efficiency of the fleet by providing real-time feedback to drivers on their eco-driving behavior and enabling fleet managers to plan optimal routes, schedules, and maintenance activities [3].

Telematics in trucks and buses is supported by a set of standards called SAE J1939, which was developed by the Society of Automotive Engineers (SAE) to standardize communication between ECUs in heavy-duty vehicles. SAE J1939 defines how these units communicate with one another and exchange information over the Controller Area Network (CAN) [4].

SAE J1939 is a key standard for telematics in trucks and buses, as it ensures compatibility and interoperability among different components and suppliers. It also facilitates the integration of telematics devices that can collect and transmit data from the vehicle to a central server or a cloud platform. SAE J1939 allows telematics devices to access a wide range of data signals from the vehicle, such as engine RPM, fuel level, battery voltage, tire pressure, brake status, fault codes, and more. [5] These signals are called Suspect Parameter Numbers (SPN) in SAE J1939, and there are more than 3500 SPNs defined. SPNs can provide valuable information about the vehicle's condition, performance, and efficiency.

The data available on the CAN network that follows the SAE J1939 protocol can be obtained from the OBD II data connector. The OBD II (On-Board Diagnostics, Second Generation) is a standard for vehicle diagnostics and emissions testing. The OBD II uses a female 16-pin (2x8) SAE J1962 connector on the vehicle [6]. The SAE J1939 uses the CAN ISO 11898 specification for the physical and data-link layers and defines a set of messages and parameters for vehicle data. The data collection from the OBD II connector is simple and non-invasive, as it does not require any modification or installation of additional devices on the vehicle. The data can be accessed by using a compatible telematics device that can connect to the OBD II port and read the SAE J1939 messages [4].

Fig. 1 illustrates the process from acquiring data from the vehicle to making it available for reading and reporting.

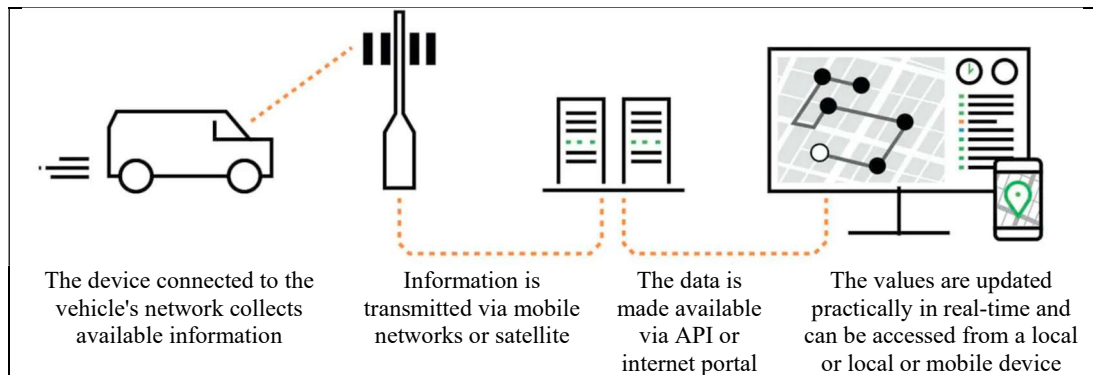


Fig. 1. How does telematics work? [7]

The use of vehicle telematics allows the capture of vehicle operation data in real-time and in large quantities. In this context, approaches related to big data and artificial intelligence, especially in machine learning, have been increasingly used, as demonstrated by the growing number of publications in recent years, especially in *machine learning*.

This paper discusses the extraction and validation of features from a database, collected from 10 trucks over 62 days, as a step in a larger project that aims to apply machine learning techniques directed to decision making for reducing fuel consumption and performing intelligent fault diagnosis. Now, however, the aim will be to use machine learning techniques to estimate the load condition of a heavy-duty vehicle using data from telematics equipment.

The paper is structured as follows: Section 2 provides a literature review, Section 3 refers to the problem description, Section 4 reports the methodology, Section 5 presents the obtained results, and, lastly, Section 6 offers conclusions and outlines future avenues of research.

2. Literature Review

Fuel consumption is one of the main factors that affect the economic and environmental performance of road freight transport. Heavy trucks consume a large amount of energy and emit significant amounts of greenhouse gases and air pollutants. Therefore, reducing fuel consumption in heavy trucks is a critical challenge for the transport sector. Telematics and machine learning are two technologies that can help to address this challenge, by providing data and solutions to improve the energy efficiency of the fleet.

A literature review was conducted to identify relevant articles on fuel consumption, heavy-duty, telematics, and machine learning. The search was performed on several databases, such as Scopus, Web of Science, IEEE Xplore, and Google Scholar. The search was limited to articles published in English from 2013 to 2022. The search resulted in several dozen articles, which were screened based on their titles, abstracts, and full texts. The articles were then grouped into three main categories: articles related to the use of data for fuel consumption reduction, articles related to driving style, and articles related to maintenance optimization.

The first category includes articles that use telematics data and machine learning methods to estimate or predict fuel consumption or emissions in heavy trucks. These articles aim to develop models that can accurately capture the relationship between vehicle data and fuel consumption or emissions and provide feedback or suggestions to drivers or fleet managers on how to improve their energy efficiency.

Some examples of these articles are:

- The use unsupervised machine learning techniques to identify the causes behind anomalous fuel consumption [8].
- A methodology for estimating emissions in real driving conditions based on on-board diagnostic data and machine learning [9].
- The uses of data collected from the SAE J1939 network and data on the characteristics of UK roads and uses Support Vector Machine (SVM), Random Forest (RF), and Artificial Neural Network (ANN) models to model the fuel consumption of trucks [10].

The second category includes articles that focus on driving style and its impact on fuel consumption or emissions in heavy trucks. These articles aim to analyze drivers' driving behavior and identify factors that affect their eco-driving performance. They also propose systems or applications that can assist drivers in improving their driving style and reducing their fuel consumption or emissions. Some examples of these articles are:

- Analyzing the influence of driving style on the total emissions of pollutants emitted by an internal combustion vehicle. The information was acquired directly from the vehicle and used automatic learning techniques to prove the direct relationship between driving style and pollutant emissions [11].
- Presenting a DrivingStyles architecture, which uses data mining techniques and neural networks to analyze and generate a classification of driving styles and fuel consumption based on the characterization of the driver capable of characterizing the degree of aggressiveness of each driver. They demonstrate the impact of driving style on fuel consumption, as well as its correlation with the greenhouse gas emissions generated by each vehicle [12].
- Propose a driving assistant that makes recommendations to reduce fuel consumption. Artificial intelligence techniques are used to issue eco-driving tips that best suit the user's profile, the vehicle's characteristics, and the road conditions [13].

The third category includes articles that deal with maintenance optimization and its effect on fuel consumption or emissions in heavy trucks. These articles aim to use telematics data and machine learning methods to monitor the condition and health of vehicles and predict failures or faults. They also propose strategies or algorithms that can optimize the maintenance plans and schedules of vehicles and reduce their downtime and repair costs. Some examples of these articles are:

- Propose a hierarchical SVM-based solution for predictive maintenance so that organizations can gain real-time decision-making capabilities. Companies can perform predictive maintenance and operations management with workflows triggered by analytical results. With this real-time, actionable knowledge, customers can make

predictive maintenance a reality based on real-time analytics across their entire field deployment [14].

- Deal with learning predictors per vehicle with the aim of predicting the level of utilization the next day and the time remaining until the next maintenance. It explores the performance of linear and non-linear models, showing that machine learning models can capture the underlying trends that describe non-stationary vehicle usage patterns [15].
- Propose a machine learning-based software/hardware system to reduce commercial truck downtime, warranty costs, and time to identify malfunctions [16].

The literature review shows that there is a growing interest in applying telematics and machine learning technologies to improve the energy efficiency of road freight transport. However, there are also some challenges and limitations that need to be addressed, such as data quality, availability, privacy, security, standardization, interoperability, scalability, and explainability. Future research directions may include developing more advanced and robust machine learning models, integrating multiple data sources and sensors, exploring the potential of edge and cloud computing, and evaluating the real-world impacts and benefits of telematics and machine learning solutions.

3. Problem description

Determining the loading state of a truck is a challenge for fleet managers, as it relies on human information, which can be inaccurate or incomplete. A promising solution is to use truck operating and location data, applied to machine learning techniques, to determine the loading state automatically and accurately.

Anonymized data from a carrier was received. It is known that the carrier does not usually transport fractional loads, and that the loads have the same volume, with variations of less than 1%. This is relevant to this study, as it ensures impartiality in the data treatment.

3.1. Database description and analysis

This study uses a database of telematics data collected from 10 diesel trucks of a transportation company. The trucks are equipped with GPS, internal sensors, and other data acquisition devices.

The data was collected at a frequency of 174 variables per second (1Hz). However, some data was missing, resulting in a loss of 0.31% of the temporal records.

The data was analyzed to identify 123 variables with their respective Suspect Parameter Numbers (SPNs). Some of these variables came from the GPS, while others came from the CAN network of the trucks.

Due to the large size of the dataset, descriptive measures of some variables were extracted, such as max, min, mean, mode, median, and standard deviation. This approach was inspired by the literature that evaluated image feature extraction for tumor diagnosis [17].

Statistics were extracted by grouping the data according to the operating state of the trucks (off, stopped with the engine running, or moving). The engine was defined as off when the rotation speed was below the minimum idle speed and the truck stopped when the speed was equal to zero km/h.

During the analysis, it was found that some data indicated the truck with rotation below the minimum idling speed and with speed greater than zero. This data was disregarded for analysis purposes.

For each operating state, the interval duration, traveled trip size, and statistics related to speed, rpm, torque percentage, percent load, latitude, longitude, odometer, total fuel consumption, and percent fuel level were extracted.

The data obtained by telematics was complemented with technical information from the trucks, supply reports [18], [19], [20], [21] and [22], and truck load condition reports. This information was used to validate the new database for estimating the volume of fuel consumed as well as the route traveled.

4. Methodology

The methodology for determining the loading state of a truck is composed of nine steps, as described below, and illustrated in Fig. 2:

1. Data validation: the data available from the truck's operation is validated with data from external sources. In this case, the estimated fuel consumption is compared to the refueling volume reported by the gas station.
2. Feature selection: the features or variables that will be used to feed the machine learning routine are selected.
3. Route definition: the study route is defined as short (0.5 to 5 km) or long (5 to 500 km).
4. Dimensionality reduction: the dimensionality of the data is reduced using PCA, an algorithm that preserves the original variance of the data. PCA starts with 2 dimensions and increases by 1 dimension until the explained variance is greater than 90%.
5. Data normalization: the data is normalized so that it has a mean of 0 and a standard deviation of 1.
6. Data split into training and test sets: the data set is split into two parts, with 75% for training and 25% for testing.
7. Application of machine learning methods: the machine learning methods are applied to the training set, using all features and with the PCA dimension that explains 90% of the variance.
8. Evaluation of results: the accuracy of each result is calculated on the test set.
9. Repeating the validation: steps '3' to '9' are repeated 10 times for each study route.

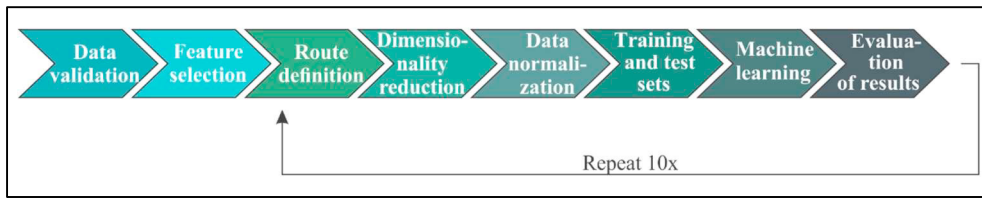


Fig. 2. Methodology for determining the loading state of a truck.

4.1. Data validation for volume and path measurement

Three methods were considered for calculating the volume of fuel consumed:

1. The variation of the fuel tank level (%*TL*). This method estimates consumption by the product between the nominal capacity (*NC*) in liters of the tank specified by the manufacturer and the variation in the percentage of the fuel level (SPN 96) during the fueling process, as described by Equation 1.

$$Vol_{k=TL} = (\%TL_{MAX} - \%TL_{MIN}) * NC \quad (1)$$

2. The accumulated diesel consumption value in liters (*TFU*). This method uses the value recorded by SPN 250 (for DAF trucks) and SPN 5054 (for Volvo trucks). The estimated consumption is given by the difference in the value indicated by the respective SPN between the end and the beginning of the operating state, as described by Equation 2.

$$Vol_{k=TFU} = TFU_{MAX} - TFU_{MIN} \quad (2)$$

3. The instantaneous fuel flow rate (*FR*). This method uses the value recorded by SPN 183. The consumed value is obtained by adding the product between the instantaneous fuel flow rate (*Q*) (SPN 183) converted to liters per second and the time (*t*) interval in seconds, as described by Equation 3.

$$Vol_{k=FR} = \sum_{i=1}^n Q_{ij}(t_i - t_{i-1}) \quad (3)$$

The first method provides a rough estimate of fuel consumption between fills, the second between runs, and the third at each captured and transmitted data.

When analyzing the data individually, it was found that episodes of small fill-ups (less than 300 liters) are more susceptible to greater divergence between the value recorded at the fuel pump and the variation in fuel level (SPN 96).

However, when accumulating the refueling throughout the analyzed time, the results indicate that, in the worst case, the maximum difference is below 9%, as illustrated in Fig. 3.

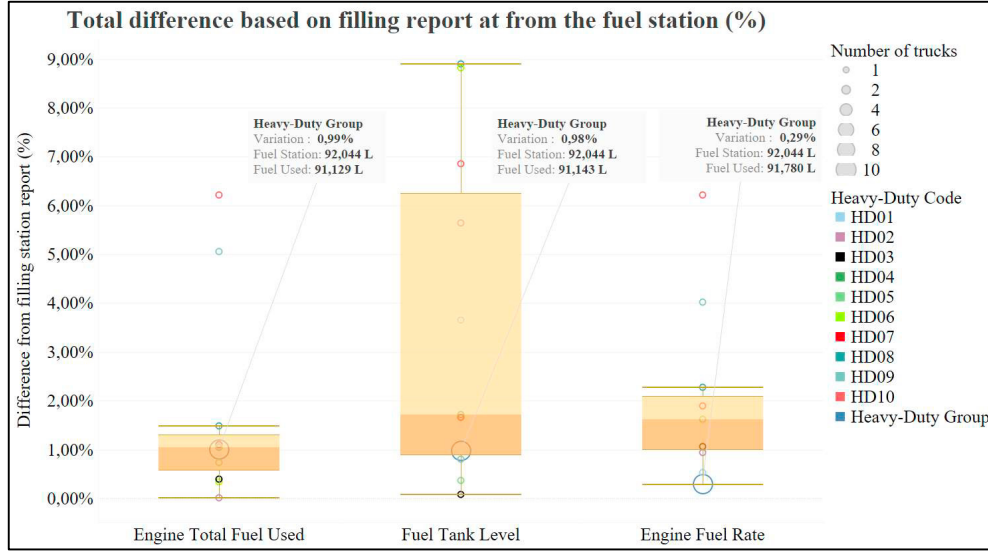


Fig. 3. Percentual difference between the methods applied to estimate the filled volume and the fueling report.

The fueling records made available by the company informed that the 10 trucks filled up 92,044 liters of diesel in the period evaluated by the study, discounting the first supply of each truck. In general, the third method of estimating consumption, estimating volume by instantaneous flow using the SPN 183, was the closest to the supply report, presenting a difference of only 264 liters (0.29%) about the 92,044 liters recorded, while the total consumption variation presented a difference of 0.99% and the method by level variation a difference of 0.98%.

Equation 4 describes the comparison between the value of the distance traveled (D) by the variation of the odometer (ΔO) (SPN 917) with the estimated displacement given by the product between the recorded speed (V) (SPN 84) converted to km/s and the admitted time (t). For trips under 300 meters, it is possible to find differences of more than 5% between the estimated value and the one registered by the odometer. For routes over 10 km, this difference was less than 0.59%, and with a tendency to reduce as the distance traveled increases.

$$\Delta O_{k=\text{odometer}} = O_{MAX} - O_{MIN} \approx D_{k=\text{speedometer}} = \sum_{i=1}^n V_i(t_i - t_{i-1}) \quad (4)$$

4.2. Inserting the relief among the relevant characteristics

The good approximation obtained by calculating the volume based on instantaneous flow (Q) (SPN 183) and the route based on speed (V) (SPN 84) made it possible to include relief features in the list of characteristics available for future analysis [23]. Since the validation of the flow variables makes it possible to calculate fuel economy (FE), as described by Equation 5, considering the relief condition:

1. ‘uphill’ – the current altitude is higher than the altitude at the previous instant.
2. ‘downhill’ – the current altitude is lower than the altitude at the previous instant.
3. ‘flat’ – the current altitude is equal to the altitude at the previous instant.

$$FE_k = \frac{D_k}{Vol_k} = \frac{\sum_{i=1}^n V_{ik}(t_i - t_{i-1})}{\sum_{i=1}^n Q_{ik}(t_i - t_{i-1})} \quad (5)$$

Where the index "k" represents uphill, downhill, flat or any relief condition.

It is important to highlight that the altitude measurement reported by the GPS is influenced by several factors besides the slope of the track, such as GPS signal quality and shaking. For this study, the possible errors of this nature

were considered negligible.

The overall average ($\overline{FE}$) of all the consumption of the 'p' sections was calculated as in Equation 6:

$$\overline{FE}_k = \frac{\sum_{j=1}^p D_{jk}}{\sum_{j=1}^p Vol_{jk}} \quad (6)$$

Where the index "j" represents each of the driving sections, and "k" uphill, downhill, flat or any relief condition.

And the difference (ΔFE) between the fuel economy on the driving stretch and the overall average was calculated, as in Equation 7:

$$\Delta FE_k = FE_k - \overline{FE}_k = \frac{\sum_{i=1}^n V_{ik} (t_i - t_{i-1})}{\sum_{i=1}^n Q_{ik} (t_i - t_{i-1})} - \frac{\sum_{j=1}^p D_{jk}}{\sum_{j=1}^p Vol_{jk}} \quad (7)$$

4.3. Feature selection

The characteristic group was defined by its creation method. Fig. 4 depicts the development of the 29 features.

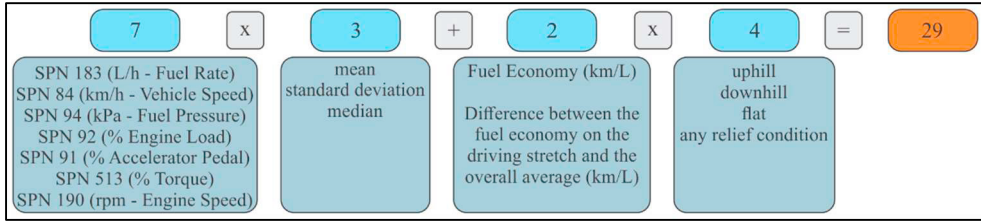


Fig. 4. Feature group creation.

4.4. Route Definition

Inspired by a study that carried out experiments on 5 km trips with low and heavy traffic [24], we use this number to separate short distances (0.5 to 5 km) and long distances (5 to 500 km). It should be noted that in the available base, there are no distances longer than 500km.

4.5. Dimensionality reduction

Principal Component Analysis (PCA) is a powerful statistical technique employed to reduce the dimensionality of a dataset effectively. It accomplishes this by projecting the data onto a lower-dimensional space while preserving as much of the underlying variation as possible. PCA achieves this by identifying and extracting the principal components, which are the fundamental directions of maximum variability present in the dataset. These principal components are derived from the eigenvectors of the dataset's covariance matrix. The associated eigenvalues of the covariance matrix quantify the extent to which each principal component explains the dataset's variance. The cumulative explained variance, a vital metric, results from summing the squared explained variances, indicating the collective variance proportion accounted for by the selected principal components [25].

4.6. Machine Learning

Normalization is a crucial preprocessing step in the realm of machine learning. It holds significance because individual features within a dataset often span varying ranges of values, potentially complicating the learning process for machine learning algorithms. The primary objective of normalization is to harmonize these disparate ranges, rendering them more conducive for effective algorithmic analysis [25].

This harmonization is achieved through a well-defined procedure: first, the mean of each feature is subtracted, followed by division by the standard deviation. This transformation ensures that all features are brought within a similar scale, enabling the machine learning algorithm to better discern and learn the intricate relationships between them. In essence, normalization lays the foundation for robust and accurate model training by mitigating the potential biases introduced by varying feature scales. [25].

To measure the performance and improvement of machine learning algorithms, it is essential to split data into at least two groups: training and test. Training data is a subset of the available data that is used to adjust the parameters of the algorithm. Test data is a subset of new, unseen data that is used to check the accuracy and generalization of the algorithm on data that it has not seen before. By dividing data into training and test sets, we can avoid overfitting, which is a situation where the algorithm performs well on the training data but poorly on the test data, indicating that it has learned the specific features of the training data but failed to capture the general patterns. Therefore, splitting data into training and test sets is a crucial step in machine learning to ensure that the algorithm can learn from data and make reliable predictions on new data [26].

The classification process used eight machine learning algorithms from the Scikit-learn library [26]:

- **AdaBoost:** This is an ensemble method that combines multiple weak learners to create a strong learner. It works by training a series of decision trees on the data, and then weighting the predictions of the trees according to their accuracy.
- **Decision Tree:** This is a non-parametric classifier that builds a tree-like structure to represent the relationships between the features and the target variable. It is a versatile classifier that can be used for a variety of tasks, but it can be sensitive to overfitting.
- **K-Nearest Neighbors (KNN):** This is a non-parametric classifier that predicts a new data point's class by identifying the k most similar data points in the training set and assigning the new data point to the majority class among its k nearest neighbors. It's user-friendly but can be computationally demanding for large datasets.
- **Logistic Regression:** This is a linear classifier that predicts the probability of a data point belonging to a particular class. It is a versatile classifier that can be used for a variety of tasks, but it can be sensitive to outliers.
- **Naive Bayes:** This is a simple probabilistic classifier that assumes that the features are independent of each other. It is a fast and easy-to-implement classifier, but it can be inaccurate for complex datasets.
- **Neural Network:** This is a machine learning model that is inspired by the human brain. It is a powerful classifier that can learn complex relationships between the features and the target variable, but it can be computationally expensive and difficult to train.
- **Random Forest:** This is an ensemble of decision trees. It works by training multiple decision trees on different subsets of the data and then combining their predictions. It is a more accurate classifier than a single decision tree, but it can be more computationally expensive.
- **Support Vector Machine (SVM):** This is a non-parametric classifier that finds a hyperplane that separates the data points into different classes. It is a powerful classifier, but it can be computationally expensive for large datasets.

5. Results

Table 1 provides an overview of route characteristics and their respective explained variances. Notably, there are more samples available for long routes compared to short ones, and the PCA dimensions and explained variances exhibited similar trends across both route categories.

Table 1. Route Characteristics and Explained Variance.

Description	Number of samples	Minimum Route (km)	Average Route (km)	Maximum Route (km)	N_PCA	Explained variance (%)
Short stretch	1016	0.501508	1.74643	4.99907	7	91.86
Long stretch	2386	5.00237	67.9266	252.684	6	91.64

Table 2 facilitates a comparative analysis of eight classification methodologies, with and without PCA, across both short and long route conditions. To enhance clarity, the most favorable outcomes have been accentuated in green.

Table 2. Accuracy Results (%) for Different Classifiers.

Description	Variables	Adaboost	Decision Tree	KNN	Logistic Regression	Naive Bayes	Neural Network	Random Forest	SVM
Short stretch	29	78.35	67.72	85.43	79.13	71.26	75.98	81.50	79.13
Long stretch	PCA 7	72.83	67.72	78.74	75.98	66.54	80.31	75.98	74.41
	29	94.30	90.12	94.97	94.64	92.96	94.14	94.97	94.97
	PCA 6	93.63	91.79	94.97	94.30	93.30	93.63	94.64	94.64

5.1. Discussion about sources of error

There are several other sources of error in measurement processes. Each SPN specifies a resolution, but this topic highlights sources of error that are at this point disregarded from the process and taken as true. As an example:

- The tank volume is provided by the manufacturer, but it may not be accurate. This is because the tank level indicator indicates that the tank is full when refueling is complete. However, the actual amount of fuel in the tank may be slightly higher or lower than the indicated value. Additionally, the density of diesel is affected by temperature, which can affect the volume measurement.
- The amount of fuel refueled was assumed to be correct, even though there are annotation errors and intrinsic measurement errors.
- The truck load was assumed to be either fully loaded or fully empty. The truck may travel with partial loads, which can be a source of error.
- The altitude (by GPS) was assumed to be accurate, even though there is an error associated with this measurement.
- Supervised machine learning methods were applied based on responses obtained manually from truck drivers.

Despite the awareness of these possible sources of error, and before seeking mechanisms that could minimize them, it was decided to evaluate the reliability of the data and the applied procedure.

6. Conclusion

The study aimed to analyze data from 10 heavy-duty vehicles provided by a heavy haul trucking company to extract relevant characteristics to be used as input variables in the study of energy saving, among other analyses, in road transport using machine learning.

When evaluating fuel saving and traveled distances it was hoped to validate the database by taking external data as reference, from the fueling report and the truck's odometer. Three estimates for the volume of diesel consumed and two for the distance traveled were presented. It is understood that the fact that they have small differences (less than 5% for consumption and 1% for distance traveled) between their results validates the data used for the estimates: percent fuel level (SPN 96), total fuel consumption (SPN 250 or SPN 5054), instantaneous fuel flow (SPN 183), odometer (SPN 917) and speed (SPN 84). Such interpretation was important to expand the understanding that the entire database was reliable.

From this, the validation of the methodology of estimating the volume by the instantaneous fuel flow and the route by the vehicle speed allowed the calculation of energy consumption in fragmented sections, classified with the help of the altitude revealed by the GPS in uphill, downhill, and flat. This result reflects the importance of data analysis for more precise management of vehicle operating conditions.

The machine learning techniques proved to be very efficient in classifying the truck's load condition, opening space for studies that can quantify the load carried only with the data present in the SAE J1939 network.

The way the database was built may have hindered the classification in small distances, so the future proposal is to modify the calculation so that it has among the small distances, all the travel starts at the large distances. This will make the sample base grow as the minimum distance traveled decreases, and perhaps it will be possible to determine the loading condition of the truck already in the first meters of travel.

Finally, this study showed that freight vehicle telematics data can be reliably analyzed to design studies that answer questions aimed at reducing energy consumption in freight transportation, among other questions, benefiting both the operation and the environment.

Acknowledgments

The authors thank CAPES, PPGMEC, UFG, and especially the Grupo Masut for their collaboration and availability.

References

- [1] Empresa de Pesquisa Energética (EPE), Agência Internacional de Energia (AIE). Transporte Rodoviário de Cargas – Brasil 2021 – Benchmarking Internacional, Brasil, Ministério de Minas e Energia, 2021.
- [2] Daley, Ryan, and Matthew Helm. 2019. Telematics and Data Science: Informing. Energy-Efficient Mobility. Golden, CO: National Renewable Energy Laboratory. NREL/SR-5400-73661. <https://www.nrel.gov/docs/fy19osti/73661.pdf>.
- [3] Martulli A, Galvin R and Ruzzenenti F (2021) Evolution of Energy and Exergy Efficiency in the European Road Freight Industry, 1978–2018. *Front. Energy Res.* 9:609077. doi: 10.3389/fenrg.2021.609077.
- [4] SAE International, “SAE J1939 Standards Collection”, https://www.sae.org/publications/collections/content/j1939_dl/ (accessed: 28/05/2023)
- [5] Alexandra Novotná. Autopi.io, “SAE J1939 Simplified: The Ultimate (2023) Guide”, <https://www.autopi.io/blog/j1939-explained/> (accessed: 31/08/2023).
- [6] Lantronix, “How to collect CAN FMS/J1939/OBD-II data with FOX3-2G/3G/4G Series”, https://cdn.lantronix.com/wp-content/uploads/pdf/AppNote_CAN_FMS_CAN_OBDII_Howto.pdf (accessed: 31/08/2023).
- [7] Movildata, “¿Qué es la telemática?”, <https://movildata.com/recursos/que-es-la-telematica/> (accessed: 31/08/2023)
- [8] Alberto Barbado, Óscar Corcho. Interpretable machine learning models for predicting and explaining vehicle fuel consumption anomalies. *Engineering Applications of Artificial Intelligence*. 2022; Volume 115, 105222, ISSN 0952-1976, <https://doi.org/10.1016/j.engappai.2022.105222>.
- [9] Rivera-Campoverde ND, Muñoz-Sanz JL, Arenas-Ramirez BdV. Estimation of Pollutant Emissions in Real Driving Conditions Based on Data from OBD and Machine Learning. *Sensors*. 2021; 21(19):6344. <https://doi.org/10.3390/s21196344>
- [10] F. Perrotta, T. Parry and L. C. Neves. Application of machine learning for fuel consumption modelling of trucks. 2017 IEEE International Conference on Big Data (Big Data). Boston, MA, USA, 2017, pp. 3810-3815, doi: 10.1109/BigData.2017.8258382.
- [11] Rivera, N.D., Molina, P.A., Bermeo, A.K., Bermeo, O.E., Figueroa, J.L. (2022). Driving Style Analysis by Studying PID's Signals for Determination of Its Influence on Pollutant Emissions. In: Rocha, A., López-López, P.C., Salgado-Guerrero, J.P. (eds) *Communication, Smart Technologies and Innovation for Society. Smart Innovation, Systems and Technologies*, vol 252. Springer, Singapore. https://doi.org/10.1007/978-981-16-4126-8_30
- [12] J. E. Meseguer, C. K. Toh, C. T. Calafate, J. C. Cano and P. Manzoni. Drivingstyles: a mobile platform for driving styles and fuel consumption characterization. in *Journal of Communications and Networks*, vol. 19, no. 2, pp. 162-168, April 2017, doi: 10.1109/JCN.2017.000025.
- [13] V. C. Magaña and M. Muñoz-Organero. Artemisa: A Personal Driving Assistant for Fuel Saving. in *IEEE Transactions on Mobile Computing*, vol. 15, no. 10, pp. 2437-2451, 1 Oct. 2016, doi: 10.1109/TMC.2015.2504976.
- [14] - Chaudhuri, Arindam & Ghosh, Soumya. (2021). Predictive Maintenance of Vehicle Fleets Using Hierarchical Modified Fuzzy Support Vector Machine for Industrial IoT Datasets. 10.1007/978-3-030-86271-8_28.
- [15] - Dena Markudova, Sachit Mishra, Luca Cagliero, Luca Vassio, Marco Mellia, Elena Baralis, Lucia Salvatori, Riccardo Loti. Preventive maintenance for heterogeneous industrial vehicles with incomplete usage data. *Computers in Industry*, Volume 130, 2021, 103468, ISSN 0166-3615, <https://doi.org/10.1016/j.compind.2021.103468>.
- [16] - Sun, Yong & Xu, Zhentao & Zhang, Tianyu. (2019). On-Board Predictive Maintenance with Machine Learning. SAE Technical Paper 2019-01-1048, 2019, doi:10.4271/2019-01-1048.
- [17] W. Nick Street, W. H. Wolberg, O. L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. *Proc. SPIE 1905, Biomedical Image Processing and Biomedical Visualization*, (29 July 1993); <https://doi.org/10.1117/12.148698>
- [18] Volvo Truck Corporation. Folha de Especificações FH Motor Motor D13C540, EU5SCR-M – POR – Version 02. (2010)
- [19] Anfavea. Tabela de Emissões Diesel Volvo. (2015)
- [20] DAF. PACCAR MX-13 engines. (2019)
- [21] DAF. Ficha Técnica XF105 FTT. (2017)
- [22] DAF. DAF Caminhões - Índices de Emissão. (2022)
- [23] Hans Jakob Walnum, Morten Simonsen. Does driving behavior matter? An analysis of fuel consumption data from heavy-duty trucks. *Transportation Research Part D: Transport and Environment*, Volume 36, 2015, Pages 107-120, ISSN 1361-9209, <https://doi.org/10.1016/j.trd.2015.02.016>.
- [24] Dimitrios Rimpas, Andreas Papadakis, Maria Samarakou, OBD-II sensor diagnostics for monitoring vehicle operation and consumption. *Energy Reports*, Volume 6, Supplement 3, 2020, Pages 55-63, ISSN 2352-4847, <https://doi.org/10.1016/j.egyr.2019.10.018>.
- [25] Alpaydin, E. (2014). *Introduction to Machine Learning*. Third Edition, MIT Press.
- [26] Pedregosa, F. and Varoquaux, G. and Gramfort, A. and Michel, V. and Thirion, B. and Grisel, O. and Blondel, M. and Prettenhofer, P. and Weiss, R. and Dubourg, V. and Vanderplas, J. and Passos, A. and Cournapeau, D. and Brucher, M. and Perrot, M. and Duchesnay, E., “Scikit-learn: Machine Learning in Python”, *Journal of Machine Learning Research*, volume 12, pages 2825-2830 (2011)